2025 IEEE International Conference on Robotics and Automation (ICRA)
May 19-23, 2025. Atlanta, USA

BUMBLE: Unifying Reasoning and Acting with Vision-Language Models for Building-wide Mobile Manipulation

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***Abstract*— To operate at a building scale, service robots must perform long-horizon mobile manipulation tasks by navigating to different rooms, accessing multiple floors, and interacting with a wide and unseen range of everyday objects. We refer to these tasks as Building-wide Mobile Manipulation. To tackle these inherently long-horizon tasks, we introduce BUMBLE, a unified Vision-Language Model (VLM)-based framework integrating open-world RGB-D perception, a wide spectrum of gross-to-fine motor skills, and dual-layered memory. Our extensive evaluation (90+ hours) indicates that BUMBLE outperforms competitive baselines in long-horizon building-wide tasks that require sequencing up to 12 skills, spanning 15 minutes per trial. BUMBLE achieves 47.1% success rate averaged over 70 trials in different buildings, tasks, and scene layouts from various starting locations. Our user study shows 22% higher task satisfaction using our framework compared to state-of-the-art VLM-based mobile manipulation methods. Finally, we show the potential of using increasingly capable foundation models to improve the system performance further. For more information, see https://robin-lab.cs.utexas.edu/BUMBLE/**

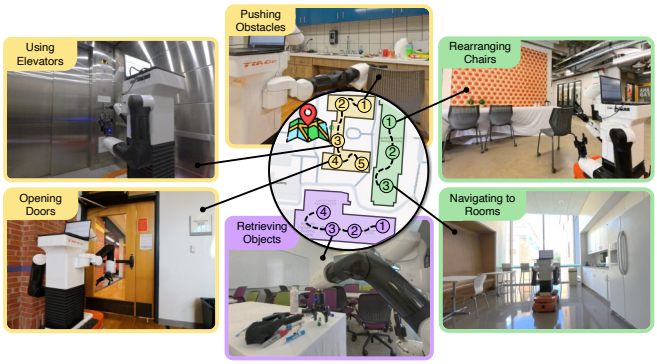


Fig. 1. **Building-wide mobile manipulation**. At a building-wide scale, mobile manipulation tasks involve sequencing multiple skills like pushing obstacles or opening doors to clear robot pathways, using elevators to reach a destination floor, rearranging chairs in the workspace, and retrieving objects. We present a framework, BUMBLE, that can solve long-horizon, spatially expansive tasks across different buildings.

I. Introduction

Autonomous service robots aiming to assist humans daily must perform mobile manipulation (MoMa) in homes, hospitals, offices, and other building-wide environments. Given a user instruction, like "*I am on a diet, but I want soda*," the robot must interpret the free-form instruction, create a high-level task plan to accomplish the task, and instantiate low-level robot commands to achieve it. To accomplish such tasks in buildings, the robot might have to traverse to a kitchen, potentially using an elevator if it is on a different floor. To use the elevator, the robot must draw on prior experiences with elevators in other buildings. The robot must also handle unexpected obstacles, such as by pushing items blocking a narrow corridor or opening doors to move in and out of the room. Finally, even after reaching the kitchen, the robot must identify an appropriate soda can from the diverse clutter often present in human-inhabited buildings, recognizing the "diet" option even if it has never been encountered before. We refer to such long-horizon and spatially expansive tasks as **Building-wide Mobile Manipulation** (see Fig. 1) and present a unified framework to address it.

Building-wide MoMa entails a series of technical challenges, including integrating complex locomotion and manipulation behaviors in buildings, perceiving objects and their context in the open world, and using past experiences to learn from mistakes reason about long-horizon strategies (example executions shown in Fig. 2). Decades of research in Task and Motion Planning (TAMP) [1–4] and recent neuro-symbolic planning [5–8] have been applied to long-horizon robotic tasks. However, these approaches require pre-defined symbolic abstractions and domain knowledge, confining their reasoning capabilities to a closed set of objects. Large Language Models (LLMs) [9–14] are capable of reasoning about open-world scenarios to create task plans. However, they lack grounded and direct reasoning through visual perception, inherently falling short in the precise geometric reasoning crucial for producing executable robot actions. Recently, Vision-Language Models (VLMs) have emerged as a promising tool for Building-wide MoMa, unlocking grounded reasoning in open-world scenes [15–23]. Some works [24–28] have demonstrated the potential of VLMs in simple MoMa tasks but lack the necessary memory and motor skills to operate at a building scale. In contrast, building-wide MoMa requires 1) an open-world perception system for reasoning about diverse objects, 2) complex motor skills to act effectively in buildings, and 3) memory for temporally extended reasoning in long-horizon task execution.

We propose to tackle Building-wide MoMa by unifying reasoning and acting through a VLM-based framework—one that can handle diverse objects with open-world perception, act effectively in building-wide situations with a broad set of gross-to-fine motor skills, and learn and adapt from past experience through memory. By seamlessly unifying these capabilities into rapidly advancing VLMs, our general framework improves as VLM backbones become better at

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2025 IEEE International Conference on Robotics and Automation (ICRA) | 979-8-3315-4139-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICRA55743.2025.11128444

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Fig. 2. **Building-wide mobile manipulation** tasks require navigating various rooms and floors while interacting with diverse objects. To solve such long-horizon tasks, BUMBLE leverages a diverse skill library enabling navigation and interaction, with parameterized skills adaptable to different scenes. At each skill execution step (circled numbers), BUMBLE uses VLM's reasoning capabilities to select the next skill (blue text) and skill parameters (**[...]**) and recover from failures (red text) to solve building-wide tasks effectively.

sophisticated perception and reasoning.

To this end, we introduce BUMBLE (**BU**ilding-wide **MoBiLE** Manipulation), a new framework that addresses all of these needs with an adaptable approach capable of reasoning and acting over new objects, tasks, and buildings. Given a task instruction in free-form language, BUMBLE reasons over the scene, past experiences, and motor skill capabilities to predict the next parameterized skill to execute. BUMBLE comprises four key ingredients: a) a **VLM** serving as the central reasoning module connecting perception, memory, and skills; b) **dual-layered memory**: short-term memory [29] to maintain robot *execution history*, and long-term memory [30] to store valuable experience and concepts from *past trials*; c) a diverse **skill library** of parameterized skills that enable the robot to navigate to different floors and rooms (e.g., `GoToLandmark[GoalImage]`, `UseElevator[Button]`), adjust the robot base for manipulation (e.g., `MoveBase[Dir.]`), and interact with objects through diverse contact-rich behaviors (e.g., `PushObjOnGround[ObjSeg., Dir.]`) present in different buildings; d) **perception system** for the VLM to not only visually reason about open-world, cluttered scenes but also process depth information necessary for embodied decisions.

Our experiments demonstrate that BUMBLE can successfully execute long-horizon building-wide tasks requiring up to 12 parameterized skills and over 15 minutes per trial (excluding VLM query time). These tasks involve navigating different floors and rooms, clearing pathway obstacles, retrieving objects, and making granular decisions, such as adjusting the base for manipulation. Notably, BUMBLE achieves a 47.1% average success rate in tasks like retrieving soda cans and rearranging chairs across three buildings. We demonstrate that BUMBLE solves previously unseen tasks involving multiple plausible solutions and operating in completely new rooms. Through a user study, we find that BUMBLE rollouts align with human preferences 22% better than prior state-of-the-art VLM-based MoMa approaches. Furthermore, we demonstrate how BUMBLE improves performance with advancing VLM capabilities. Finally, we thoroughly analyze failures, guiding future research.

II. Related Work

Pioneer research on **mobile manipulators** [31–35] performed tasks in human environments, which naturally occur in buildings [36–38]. Early work in whole-body control [39–43] and motion planning [44–48] laid the groundwork for low-level control in building-level mobile manipulation, but did not address reasoning required in long-horizon building-wide tasks. Task and motion planning (TAMP) [1–4, 49] integrates reasoning for task and motion generation for long-horizon tasks. However, it is confined to a fixed, preprogrammed set of object categories and scenes, limiting its application to buildings with open-world objects.

Recently, **foundation models** [50] have been leveraged for mobile manipulation. LLMs excel at semantic reasoning for planning [51, 52], but their predictions are not grounded in the scene, requiring additional machinery to match their plans to executable robot actions [9, 11, 53, 54]. VLMs have been shown to be a promising framework for unifying and ground reasoning for tabletop manipulation [15–17, 26], navigation [19–22, 55], and mobile manipulation [24, 25, 27, 28, 56]. However, prior methods lack a diverse skill repertoire (navigation and manipulation, coarse and fine

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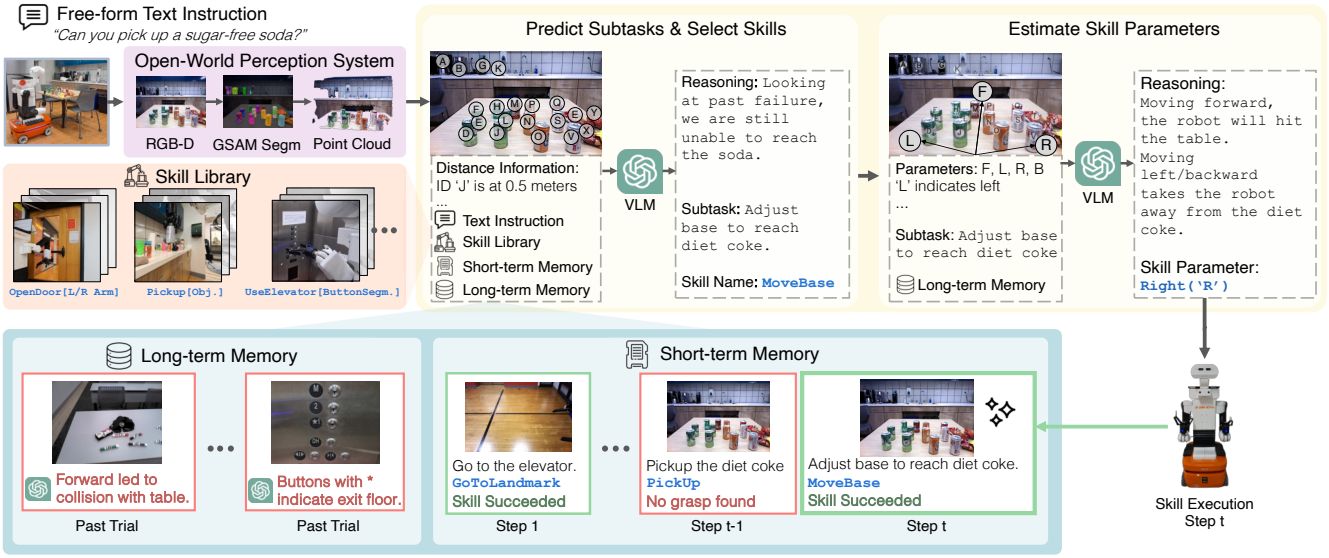


Fig. 3. **BUMBLE architecture.** Given a free-form text instruction, skill library (*top Left*), and short- and long-term memory (*bottom middle and left*), BUMBLE iteratively perceives the environment through onboard RGB-D sensors (*top left*), predicts parameterized skills (*top middle and right*), and executes them in the environment. The skill and its parameters are executed and stored in short-term memory for iterative prediction (*bottom right*).

motion) and long-horizon memory-based reasoning capability, both of which are necessary to perform building-wide mobile manipulation tasks. BUMBLE expands today's VLM-based robotic systems: it uses VLMs to unify semantic and geometric reasoning for manipulation and navigation at the building-wide scale, enabling it to leverage current and future advances in VLM models. Our framework integrates perceptual and motor skills and long and short-term memory to obtain efficiency and generalization to diverse objects and scenes in different buildings.

All aforementioned prior work in mobile manipulation assumes the navigation path is either obstacle-free or that obstacles can be navigated around. In **interactive navigation** [57, 58] however, obstacles may be moved or manipulated—such as pushing away path-blocking boxes, opening doors, or pressing elevator buttons—to reach a destination. Prior methods tackled this with motion planning [59–61] and learning [62–64] but focused solely on the interactive navigation problem using only geometric information. In contrast, BUMBLE integrates both semantic reasoning (e.g., avoiding pushing delicate objects) and geometric reasoning (e.g., avoiding object collisions while pushing an obstacle) into a unified method for mobile manipulation.

III. BUMBLE: VLM-Based Building-Wide MoMa

BUMBLE is a new framework that integrates perception, motor skills, and memory in a VLM-based solution for high-level reasoning and low-level control of a mobile manipulator for building-wide tasks. In this section, we describe the key technical blocks of BUMBLE (see Fig. 3): (a) capabilities for open-world perception, a library of diverse skills to act in the physical world, and short and long-term memory for on-the-fly adaptation and learning from past mistakes; and (b) a VLM-based decision-making module to reason and ground the text instruction in the current scene and predict the next parameterized skill to execute.

A. Perception System, Skill Library, and Memory

a) Open-World Perception System: Robots must perceive and localize diverse open-world objects to operate effectively in buildings. We address this using a perception system with a robust segmentation model, Grounded-SAM (GSAM) [65–67], for segmenting foreground (i.e., interactable) objects (Fig. 3, purple). We use GSAM instead of directly querying the VLM because segmentation models offer pixel-level accuracy, allowing for more precise localization and manipulation. After obtaining the object masks for the scene, we calculate the object point cloud by back-projecting depth images and determine the precise distance between the robot and the detected objects. The object proximity information enables BUMBLE to estimate the correct skill for manipulation in building environments, while the object point cloud enables precise manipulation using motor skills.

b) Skill Library: To overcome the challenges of building-wide MoMa, BUMBLE requires a diverse skill library allowing for high-level abstract behaviors like navigating to a room, down to more fine-grained behaviors like adjusting the base for manipulation (Fig. 3, yellow; examples in Fig. 2, 4). The skill library must also support using building-specific fixtures like elevators and perform manipulation skills like pushing objects and picking up items, addressing interactive navigation challenges [47, 58]. To this end, we construct a skill library to which new skills can easily be added and chosen by the VLM. Our library includes skills such as `GoToLandmark`, `NavigateNearObj`, `MoveBase`, `Pickup`, `PushObjOnGround`, `OpenDoor`, `CallElevator`, and `UseElevator`. The parameters of each skill are based on object or robot configurations, except

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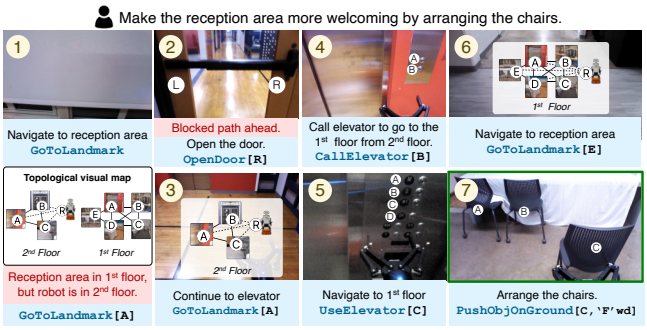


Fig. 4. **Execution trace for a user instruction.** Each decision step shows the image observation with grounded skill parameters as markers, the skill name (blue text) and parameter executed, and a brief language description of the step (black) for readability. In trying to execute some skills, the robot fails (red text), in which case the VLM adaptively predicts the next skill.

for `GoToLandmark`. It uses a topological visual map [68] of the building, with landmark images as nodes and 2D occupancy maps to generate trajectories between them (See Fig. 4). With the library of parameterized skills and VLM's reasoning capabilities, BUMBLE can integrate semantic and geometric reasoning at the task level by predicting the subtask and skill and the robot's motion by estimating skill parameters to accomplish building-wide tasks.

c) Memory: Building-wide MoMa is inherently long-horizon, requiring the robot to track its state-action history. Moreover, the robot must recover from failed skill executions whenever possible, such as by moving obstacles or adjusting the base after a failed grasping attempt. To enable these recovery behaviors, the robot must store and analyze prior failed executions from the current trial when making future predictions. To address this, BUMBLE maintains **short-term memory** (Fig. 3, blue). The short-term memory stores the scene image, subtask, skill name, parameter, and the system-detected execution result (success/failure) at each prediction step for the *current execution trial*. This allows the VLM to reason over the entire execution history when predicting its next skill, crucial for long-horizon building-wide tasks.

The VLM may make reasoning errors, such as failing to consider object collisions with the environment, which could, for example, result in pushing an obstacle into a wall. To allow BUMBLE to learn from these mistakes in reasoning and reduce them [69], BUMBLE maintains a **long-term memory** of prediction failures collected previously. Erroneous VLM predictions, flagged by a human operator, are stored together with the decision context (user instruction, scene image, predicted subtask, skill name, and predicted parameters) and a VLM text analysis of the failure reason, serving as lessons for improving future predictions and reducing the dependency on human annotations.

B. VLM-based Decision Making

BUMBLE factorizes each decision-making step into two sequential steps (Fig. 3, yellow box). First, BUMBLE exploits a VLM to perform perceptual and memory-based reasoning for predicting the next **subtask** to solve (e.g., "*go to a place where you can find a soda*") and for selecting the **skill** to handle it (e.g., `GoToLandmark`). Then, BUMBLE queries the VLM to estimate the best **skill parameters** (e.g., `goal image of kitchen`) and instantiates the corresponding motor skill execution.

a) Subtask prediction and skill selection: First, BUMBLE needs to infer the next step of the task and what skill to use to solve it. To that end, it integrates information about i) the language task instruction, ii) the description of robot skills, iii) the current scene (RGB) and robot state (location in the building), and iv) execution history (short-term memory) and failures from prior trials (long-term memory) into a *unified multimodal VLM query*.

The VLM must also perceive the object distance information to infer the object's reachability, which is necessary for manipulation. Naively prompting the VLM with depth images or point clouds does not yield desirable results because VLMs are not trained on those modalities. To allow the VLM to make informed decisions, BUMBLE implements a multimodal Set-of-Mark (SoM) [70] prompting: foreground objects are marked with IDs in the input RGB image, and their distances to the robot, calculated using the perception system (Sec. III-A), are passed in the prompt.

BUMBLE employs the resulting multimodal prompt to query the VLM-backbone about the next step in the task and the best-suited skill to solve it, requesting intermediate reasoning with Chain-of-Thought (CoT) [71] for better accuracy. With this VLM-backed procedure, BUMBLE can reason about long-horizon tasks, eliminating the need for a classical planner's computationally heavy search.

b) Skill parameter estimation: After predicting the subtask and skill to execute, BUMBLE predicts the parameters needed to correctly instantiate the skill and generate a sequence of low-level robot commands. These parameters must be contextualized and grounded in the current scene for informed decision-making. To that end, BUMBLE creates a multimodal VLM query that includes i) the predicted subtask, ii) memory of previously made mistakes in predicting the skill parameters, and iii) an SoM-based visual grounding of the possible skill parameters with explanations of each marker (example in Fig. 4). For example, for the `MoveBase` skill, the possible parameters (forward, left, and right) are marked in the RGB image as arrow endpoints. For object-centric skills, the candidate parameters are generated by querying the GSAM with a skill-specific prompt (e.g., find "Buttons" for `UseElevator`). BUMBLE queries the VLM to choose the most promising skill parameter using CoT prompting for better reasoning about the subtask, scene, and past experiences. The execution outcome is stored in the short-term memory for iterative prediction.

IV. Experimental Evaluation

To systematically evaluate the performance of BUMBLE, we benchmark **three** very long-horizon tasks up to 12 skills long: *Retrieving marker*, *Retrieving diet soda can* and *Rearranging chairs*. To test the system's language understanding capabilities, the tasks are specified using free-form text

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TABLE I

SUCCESS RATE (%) FROM 10 TRIALS IN BUILDING-WIDE TASKS

	Retrieve marker			Retrieve soda can			Rearrange chairs	Avg.
Building	*B1*	*B2*	*B3*	*B1*	*B2*	*B3*	*B1*	(%)
IM [72]	10	–	–	0	–	–	10	6.7
COME [28]	40	30	40	**40**	30	30	40	35.7
BUMBLE (Ours)	**60**	**40**	**50**	**40**	**50**	**40**	**50**	**47.1**

instructions, and the evaluations are averaged over three different task specifications for each task. For instance, in the retrieving marker task, we use variations such as "*I want to color the grass in my drawing. Can you get me a marker?*", "*I need to color some hearts. Can you get a marker for that?*", and similar prompts.

We test BUMBLE's building-wide MoMa capabilities by initializing the agent on different floors (possibly requiring elevator use), with randomized obstacles along the robot's path, such as closed doors, chairs, wet floor signs, and cardboard boxes. BUMBLE must reason geometrically (e.g., selecting the correct arm to open a door or direction to push an obstacle) and semantically (e.g., avoiding wet floor signs).

We evaluate open-world generalization with multiple target objects per task—different brands of diet soda and markers, in random positions under varying degrees of clutter (5-25 distractor objects). We test building-level generalization across three university buildings of different layouts, visual appearance, and room structures. Results are measured using 10 trials per building per task (totaling 90+ hours of evaluations). Since BUMBLE demonstrated consistent performance on the first 2 tasks across the 3 buildings, we only evaluated our third task, rearranging chairs, in one building.

We compare BUMBLE with two baselines: **Inner-Monologue (IM)** [72] that reasons using only a language scene description without RGB images or long-term memory, and **COME** [28], which reasons using images, like BUMBLE, but without long-term memory. Both methods were originally unsuitable for building-wide tasks due to a limited skill library and lack of memory, so we extended them with BUMBLE's diverse MoMa library and short-term memory for fair comparison. We instantiate BUMBLE with GPT-4o, which we find to perform best empirically. All methods use the same VLM and skill library, resulting in similar task execution times. We gather long-term memory experiences, consisting of reasoning errors made by VLM, before evaluations and do not update them later. These experiences consist of up to 3 examples per skill.

Finally, in our evaluation, we complement the full-fledged long-horizon building-wide MoMa experiments by only assessing parameter estimation for given skills in predefined scenes. To that end, we manually collected an offline dataset (`OfflineSkillDataset`) of 120 images for 3 different skills and annotated them with ground truth parameters.

In our evaluation, we investigate the following questions:

1) How does BUMBLE perform compared to other methods when addressing building-wide tasks? Table I summarizes the results of our experiments. Inner Monologue (IM) performs poorly compared to COME and BUMBLE, indicating that multimodal prompting with visual information is critical to making the right decisions. Due to its poor performance, we restrict the evaluation of IM to that first building. BUMBLE scores the best in all tasks and buildings and outperforms COME by 12.1% on average, owing to learning from past mistakes. To directly compare the reasoning capability of these approaches over a diverse but fixed set of scenarios, we test each method against the ground truth skill parameters in `OfflineSkillDataset`. BUMBLE achieves 80.2% success, outperforming COME (72.6%) and Inner-Monologue (61.7%), suggesting that BUMBLE's improvements in single-step skill parameter prediction help drive more successful executions in long-horizon tasks.

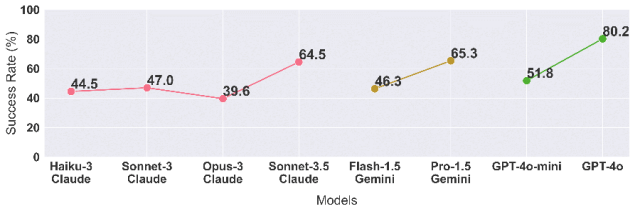


Fig. 5. **Success rate (%) of VLMs in predicting skill parameters.** The models are arranged with increasing capabilities, measured per the Vision Arena [73], from left to right in *each model series*.

2) How well does BUMBLE scale with increasing VLM capabilities? In BUMBLE, we sought to unify reasoning with a VLM backbone to leverage advances in this fast-paced field. For that, BUMBLE would have to work well with different VLMs and demonstrate improving capabilities with more powerful versions. Fig. 5 includes the results of our evaluation of BUMBLE using Claude (Red), Gemini (Yellow), and GPT-4o (Green) model series as VLM-backbone on the `OfflineSkillDataset`. BUMBLE works well with any VLM, and its performance with each model scales with the VLM capabilities, highlighting the potential of BUMBLE to leverage next-generation foundation models.

3) What are the common failure modes in BUMBLE? We analyze and break down the failure cases observed during the evaluation of BUMBLE in the three long-horizon tasks and present the results in Fig. 6. Several of our errors (10/38) are caused by sensor failures (bad depth values, lidar failures) or GSAM segmentation errors, but most of them can be attributed to VLM reasoning mistakes, e.g., picking the wrong object, wrongly adjusting the base, or failing to solve an interactive navigation problem [47]. We notice that these errors occur more often when BUMBLE queries the VLM to localize markers demanding precise spatial understanding, such as elevator buttons, which frequently leads to pressing an incorrect one. The VLM is also more likely to pick an incorrect object when there are many distractors (20-25, 38.9%) than a few (5-10, 10.0%), indicating limitations when reasoning with clutter, possibly due to the presence of numerous markers.

During evaluations, we noticed a surprising emergent behavior: BUMBLE overcomes sensor failures using its

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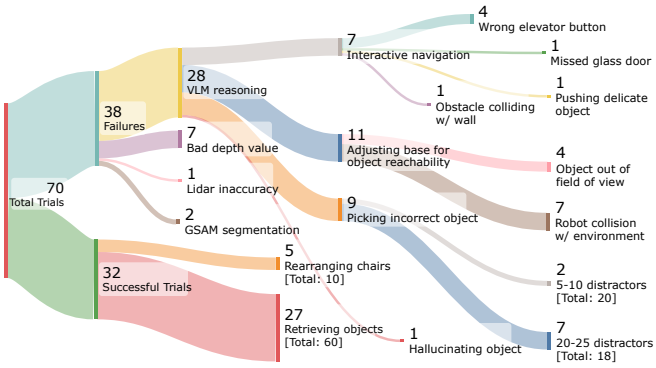


Fig. 6. **Success and failure categorization from 70 trials.** Most failures involve VLM reasoning, such as predicting actions that lead to a collision with the environment, picking up the wrong object, especially when there are 20-25 distractors, or pressing the wrong elevator button.

diverse skill library. We observed that BUMBLE learns to use `MoveBase` to a) reposition towards the elevator buttons and b) approach an obstacle (chair), both after failed manipulation attempts due to NaN depth sensor values.

4) How well does BUMBLE align to human judgement? To analyze BUMBLE's alignment with human judgment, we test two completely unseen tasks without modifying the prompts. Both tasks involve navigating to a completely new (shower) room: a) "*I spilled water on the floor. Can you pick up something to clean?*", b) "*I dropped my phone in water and dried it off with a napkin, but it's still not working. Can you pick up something to get rid of all the moisture?*". We then collect images of the execution using BUMBLE and COME and annotate them with the task description and the parameterized skill selected by each one. We ask 10 participants to annotate each trajectory by selecting the description that best fits the result of the robot's last decision. Participants choose from five categories: **1**: Irrecoverable failure causing permanent damage to the robot or environment, **2**: Recoverable failure without lasting damage, **3**: Ill-specified decision, **4**: Sub-optimal task completion, **5**: Task completion that fully satisfies the user request. Participants respond via an online form, with questions shuffled and mixed between decisions made by COME and BUMBLE.

Fig. 7 summarizes the results of our survey. BUMBLE achieves a Likert rating of 3.7 out of 5.0. Compared to COME (Avg. rating: 2.6), BUMBLE has 22% points less irrecoverable and 12% points less recoverable failures. Participants rate BUMBLE as suboptimally completing the task 33% of the time, such as successfully picking up an object, but not the one most suitable on the scene. We hypothesize this is due to greedy plans generated by VLMs, which often pick up the nearest, most visible object. Incorporating multi-step reasoning using VLMs to overcome greedy behavior is an exciting future direction for BUMBLE.

5) What is the contribution of CoT and SoM to BUMBLE? We ablate two components of the BUMBLE VLM interface: Chain-of-Thought reasoning and Set-of-Mark prompting to measure their effect on performance. Not using CoT decreases performance by 19.5% points,

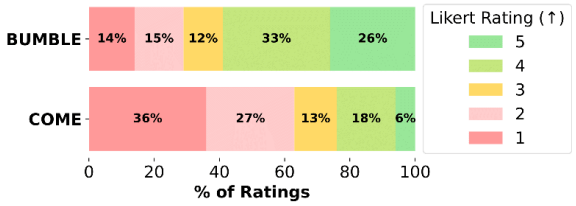


Fig. 7. **Human ratings on task completion and user satisfaction** (10 participants each evaluating 10 robot trials per method.) on a 5-point Likert scale. BUMBLE outperforms COME in aligning with human preferences.

down to 60.7%, underscoring the importance of having intermediate reasoning steps for decisions in building-wide tasks. Removing SoM and instead using the VLM to predict a desired object description in natural language, then asking GSAM to segment this object [26], decreases performance by 31% points, highlighting GSAM's difficulty in segmenting specific object instances such as "*Diet* Dr. Pepper" when a non-diet Dr. Pepper can is present.

V. Conclusion

We demonstrate that a VLM acting as a central reasoning module equipped with a diverse skill library, memory, and open-world perception system can effectively solve building-wide mobile manipulation tasks. Comprehensive ablations and experiments highlight BUMBLE's potential as a stepping stone for achieving autonomous service robots in buildings.

Limitations and Future Work: In BUMBLE, adding long-term memory experiences in the VLM's context can become computationally intractable with time; using a dynamic retrieval process to extract only relevant experiences or compressing these experiences in learnable weights [74] can be explored in future work. Furthermore, BUMBLE assumes access to landmarks with goal images in the `GoToLandmark` skill, which requires human effort in collecting the landmarks. Autonomous exploration with landmark frame selection can be explored to offload human effort. While BUMBLE easily interfaces with semantically meaningful skills, adding skills or parameters not easily expressed via language [75, 76] requires additional machinery for video conditioning or specialized weights to allow such interfacing. Lastly, fully integrating robots within buildings requires socially aware autonomy, such as social navigation [77, 78]. Extending BUMBLE framework to support these capabilities is an exciting future direction.

VI. Acknowledgments

We thank Shivin Dass for developing the robot infrastructure. We thank Ben Abbatematteo, Shuijing Liu, and Mingyo Seo for their feedback on the manuscript. We thank all UT Austin Robot Perception and Learning Lab and Robot Interactive Intelligence Lab members for their invaluable feedback. This work was partially supported by the National Science Foundation (FRR2145283, EFRI-2318065), the Office of Naval Research (N00014-22-1-2204, N00014-24-1-2550), the Army Research Lab (W911NF-25-1-0065), and the DARPA TIAMAT program (HR0011-24-9-0428).

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